

MyDesign Portfolio

Element H Initial Template

Element H: Prototype Testing and Data Collection Plan

Testing Details (H1, H2, H3, H4)

	requirement	testing setup and procedure	test cases and number of trials	expected range of results	pass/fail criteria, goal
1	The prototype must be able to be placed down on a flat surface and when free standing, there is no sliding or shifting.	1) Hold the prototype and place it down on a flat surface on the side opposite to the socket blocks. 2) When placed down, leave the holder alone for 30 seconds. 3) Note any changes in position that happen. 4) Repeat the test on wood, metal, plastic, and rubber.	a) Perform a test on wood 10 times. b) Perform a test on metal 10 times. c) Perform a test on plastic 10 times. d) Perform a test on rubber 10 times.	Parts of the prototype that may change: - All parts should remain unchanged - Parts detach - Wear down.	Fail: The socket holder shifts more than 0.15 inches in any direction on the x and y axis relative to the surface during the duration of the test. Pass: The socket holder does not shift any more than 0.15 inches in any direction on the x and y axis relative to the surface. Goal: The socket holder does not shift in any direction on the x and y axis relative to the surface.

2	mass of prototype < ?	1)			
3	The prototype must be able to hold on a wall indefinitely with the weight of the sockets on it	1) Mount the prototype with all pegs onto a pegboard with the same measurements as the one the stakeholder uses with the required amount of mass on the prototype 2) Do not interfere with the test by touching the prototype and wait 10 minutes 3) Take the prototype off and remount it to the walls 4) Repeat the process 2 more times	a. Perform the 10 minute long test on the pegboard 3 times.	Parts of the prototype may: -remain unchanged -snap -shift -detach	Fail: The prototype falls off the wall or breaks in any way during any of the 3 trials Pass / Goal: During the trials the prototype stays on the wall and doesn't break
4	Each prototype must be able to hold 10 of the sockets	1) Put sockets on the prototype being within the ½ inch, ¾ inch and ¼ inch range 2) Ten of which should remain on the prototype for 5 minutes	Perform the test one time with each of the 3 socket types	Parts of the prototype may: -remain unchanged -snap -shift	Fail: The prototype is unable to hold 10 of the sockets mentioned before at once Pass / Goal: The prototype is capable of holding 10 of the sockets mentioned before at once without failure

Mr.Reid

Works in an Auto Shop

Suggested engraving numbers on the holder that relates to the size of the socket that belongs there. Makes it better to organize it.

Alireza

Works in an Auto Shop, friends of Antonio who is the lead stakeholder.

When reviewing photos of the design he recommended making sure that we secured the metal pegs with some sort of epoxy. He also said not to use a cheap material when printing the project due to the metal pegs having the ability to fracture and cause problems in the plastic structure of the holder if too much pressure is applied to the pegs.